Dot product is an inner product for $\mathbb{R}^n$ but not the only inner product for $\mathbb{R}^n$

Ex: $((v_1, v_2), (w_1, w_2)) = v_1 w_1 + 6 v_2 w_2$

① $v = \begin{bmatrix}\alpha \\ \beta\end{bmatrix}$ $\alpha, \beta \in \mathbb{R}$

$g(v,v) = \underbrace{\alpha^2}_{\geq 0} + \underbrace{6\beta^2}_{\geq 0} \geq 0$ ✓

② $g(v,v) = \underbrace{v_1^2}_{\geq 0} + \underbrace{6v_2^2}_{\geq 0} = 0$ so $v_1, v_2 = 0$

③ $g((\alpha,\beta),(\gamma,\delta)) = \alpha\gamma + 6\beta\delta$ $g((\gamma,\delta),(\alpha,\beta)) = \gamma\alpha + 6\delta\beta$ ✓

④ $g((\alpha,\beta)+(\gamma,\delta),(\mu,\lambda)) = g((\alpha+\gamma,\beta+\delta),(\mu,\lambda)) = (\alpha+\gamma)\mu + 6(\beta+\delta)\lambda$

$g((\alpha,\beta),(\mu,\lambda)) + g((\gamma,\delta),(\mu,\lambda)) = \alpha\mu + 6\beta\lambda + \gamma\mu + 6\delta\lambda$

$= (\alpha+\gamma)\mu + 6(\beta+\delta)\lambda$ ✓

⑤ $g(\alpha v, w) = \alpha v_1 w_1 + 6\alpha v_2 w_2$ ✓

$\alpha g(v,w) = \alpha(v_1 w_1 + 6 v_2 w_2)$

Ex recall that $C^\infty(\mathbb{R} \to \mathbb{R}$ is a vector space over $\mathbb{R}$

$P(f,g) := \int_{-\infty}^{\infty} f(x)g(x)\,dx$

① $P(f,f) = \int_{-\infty}^{\infty} (f(x))^2\,dx$ $(f(x))^2 \geq 0\ \forall_x$ so $\int_{-\infty}^{\infty} (f(x))^2\,dx \geq 0$

② $P(f,f) = \int_{-\infty}^{\infty} (f(x))^2\,dx = 0$, $\int_{-\infty}^{\infty} (f(x))^2\,dx \geq 0$, if $= 0$ $f(x) = 0$

③ $P(f,g) = \int_{-\infty}^{\infty} f(x)g(x)\,dx = \int_{-\infty}^{\infty} g(x)f(x)\,dx = P(g,f)$

④ $P(f+h,g) = \int_{-\infty}^{\infty} (f(x)+h(x))g(x)\,dx = \int f(x)g(x)\,dx + \int h(x)g(x)\,dx$
$= P(f,g) + P(h,g)$

⑤ $P(\alpha f,g) = \int \alpha f(x)g(x)\,dx = \alpha \int f(x)g(x)\,dx = \alpha P(f,g)$

this innerproduct on $C^{\infty}(\mathbb{R} \to \mathbb{R})$ is called the "L^2-inner product"

Thrm inner product non-cannonicity

given finite dimensional vector space V over $\mathbb{R}$ and an inner product

$P: V \times V \to \mathbb{R}$, there is a basis for V s.t. the formula for P in coords is the dot product

$$P\left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}\right) = x_1y_1 + \dots + x_ny_n$$

↳ the dot product is basis dependent